# 12 Lecture 12

We have already seen the Cauchy-Schwarz inequality for square integrable random variables $X, Y$:

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2] \cdot \mathbb{E}[Y^2], \tag{12.1}$$

which was used to prove that the correlation $\rho(X, Y) \in [-1, 1]$. In this lecture, we will look at more inequalities related to the expectation of random variable as well as some theorems/limit theorems that will be useful in the second part of the course.

Terminology: Whenever an equality, event, limit etc. is true with probability 1, we say that it happens *almost surely.*

## 12.1 Inequalities

**Theorem 12.1** (Markov's inequality)**.** *Let $X$ be a nonnegative random variable, and let $t > 0$. Then*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

*Proof.* Let $X$ be a nonnegative random variable and let $t > 0$. Observe that

$$X \geq t\,\mathbf{1}_{\{X \geq t\}},$$

where $\mathbf{1}_{\{X \geq t\}}$ denotes the indicator of the event $\{X \geq t\}$. Taking expectations on both sides and using linearity of expectation, we obtain

$$\mathbb{E}[X] \geq t\,\mathbb{E}\big[\mathbf{1}_{\{X \geq t\}}\big] = t\,\mathbb{P}(X \geq t).$$

Rearranging gives

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t},$$

which completes the proof. □

**Example 1 (finite expectation implies finite random variable)**
Assume that $X$ is a random variable with values in $[0, +\infty]$ such that $\mathbb{E}[X] < \infty$. Let's show that $X < \infty$ almost surely.

For $n \in \mathbb{N}$ we write due to Markov's inequality

$$\mathbb{P}(X \geq n) \leq \frac{\mathbb{E}[X]}{n} \implies 1 - \frac{\mathbb{E}[X]}{n} \leq \mathbb{P}(X < n).$$

By sending $n \to +\infty$, we get

$$1 \leq \lim_{n \to +\infty} \mathbb{P}(X < n) = \mathbb{P}(X < \infty) \leq 1,$$

hence $\mathbb{P}(X < \infty) = 1$.

**Lemma 12.2** (Chebyshev's Inequality)**.** *Let $X$ be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. Then for any $t > 0$,*

$$\mathbb{P}\big(|X - \mu| \geq t\big) \leq \frac{\sigma^2}{t^2}.$$

*Proof.* Let $X$ be a random variable with mean $\mu$ and variance $\sigma^2$. Notice that

$$|X - \mu| \geq t \quad \implies \quad (X - \mu)^2 \geq t^2.$$

Since $(X - \mu)^2 \geq 0$, we can apply Markov's inequality to the nonnegative random variable $(X - \mu)^2$:

$$\mathbb{P}\big(|X - \mu| \geq t\big) = \mathbb{P}\big((X - \mu)^2 \geq t^2\big) \leq \frac{\mathbb{E}[(X - \mu)^2]}{t^2} = \frac{\sigma^2}{t^2},$$

as we wanted. □

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Let's put $Y = 1$ in (12.1). This gives $(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2]$. A more general version of this corollary of Cauchy-Schwarz' inequality is *Jensen's inequality.* Indeed, it suffices to choose $\varphi(x) = x^2$ in the theorem below.

**Theorem 12.3** (Jensen's Inequality)**.** *Let $X$ be a random variable and let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex function such that $\mathbb{E}[|X|] < \infty$ and $\mathbb{E}[|\varphi(X)|] < \infty$. Then*

$$\varphi\big(\mathbb{E}[X]\big) \leq \mathbb{E}[\varphi(X)].$$

*Proof.* Let $x_0 = \mathbb{E}[X]$. Since $\varphi$ is convex, there exists a slope $m \in \mathbb{R}$ such that[3] for all $x$,

$$\varphi(x) \geq \varphi(x_0) + m(x - x_0).$$

Applying this to the random variable $X$, we have almost surely

$$\varphi(X) \geq \varphi(\mathbb{E}[X]) + m(X - \mathbb{E}[X]).$$

Taking expectations on both sides and using linearity of expectation gives

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X]) + m\left(\mathbb{E}[X] - \mathbb{E}[X]\right) = \varphi(\mathbb{E}[X]).$$

This completes the proof. □

## 12.2 Theorems involving the expectation, limit theorems

**Lemma 12.4** (Fatou's Lemma)**.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of nonnegative random variables. Then*

$$\mathbb{E}\Big[\liminf_{n\to\infty} X_n\Big] \leq \liminf_{n\to\infty} \mathbb{E}[X_n].$$

**Example 2 (liminf of sets and inequality)**
Assume that $(A_n)_{n\in\mathbb{N}}$ is a sequence of events and $X_n = \mathbf{1}_{A_n}$. Fatou's lemma gives

$$\mathbb{E}[\liminf_{n\to+\infty} \mathbf{1}_{A_n}] \leq \liminf_{n\to+\infty} \mathbb{E}[\mathbf{1}_{A_n}] = \liminf_{n\to+\infty} \mathbb{P}(A_n).$$

We observe that the lim inf inside the expectation can be 0 or 1 depending on what the $A_n$'s are. In particular,

$$\liminf_{n\to+\infty} \mathbf{1}_{A_n}(\omega) = \begin{cases} 1, & \text{if } \omega \in \text{eventually in all } A_n\text{'s}, \\ 0, & \text{otherwise.} \end{cases}$$

We observe that

$$\omega \in \text{eventually in all } A_n\text{'s} \quad \Leftrightarrow \quad \omega \in \bigcup_{n=1}^{+\infty} \bigcap_{k\geq n} A_k =: \liminf A_n.$$

Therefore, $\liminf_{n\to+\infty} \mathbf{1}_{A_n} = \mathbf{1}_{\liminf A_n}$ and

$$\mathbb{P}(\liminf A_n) \leq \liminf_{n\to+\infty} \mathbb{P}(A_n).$$

If $A_n$ is an increasing sequence of sets ($A_n \subseteq A_{n+1}$ for all $n$), then by the continuity of the probability measure, the right hand side is $\mathbb{P}(\cup_{n=1}^{+\infty} A_n)$, which is equal to the left hand side (**why?**). This is a case of equality in Fatou's lemma.

[3] By convexity of $\varphi$, for all $x \neq x_0$ the slope of the secant line through $(x_0, \varphi(x_0))$ and $(x, \varphi(x))$ satisfies $\frac{\varphi(x)-\varphi(x_0)}{x-x_0}$. Define

$$m_1 := \sup_{x<x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0}, \qquad m_2 := \inf_{x>x_0} \frac{\varphi(x) - \varphi(x_0)}{x - x_0}.$$

Convexity guarantees that $m_1 \leq m_2$, so any $m$ satisfying $m_1 \leq m \leq m_2$ defines a line $y = \varphi(x_0) + m(x - x_0)$ that lies below the graph of $\varphi$ for all $x$, i.e., $\varphi(x) \geq y$ for all $x$.

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Question: Can we have strict inequality? $\mathbb{P}(\liminf A_n) < \liminf_{n\to+\infty} \mathbb{P}(A_n)$.

The answer is yes. One example is to consider $\Omega = \{0,1\}$ with $\mathbb{P}(\{0\}) = 1/2 = \mathbb{P}(\{1\})$ and the sets $A_n = \begin{cases} \{0\}, & \text{if } n \text{ is even,} \\ \{1\}, & \text{if } n \text{ is odd.} \end{cases}$ Then, $\liminf A_n = \emptyset$, but $\mathbb{P}(A_n) = 1/2$ and $\liminf \mathbb{P}(A_n) = 1/2$.

The reason why we have equality when $(A_n)$ is increasing is also because of the *Monotone convergence theorem.*

**Theorem 12.5** (Monotone Convergence Theorem)**.** *Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of nonnegative random variables such that*

$$X_1 \le X_2 \le X_3 \le \cdots \quad \textit{almost surely.}$$

*Suppose that*

$$X := \lim_{n\to\infty} X_n \quad \textit{exists almost surely.}$$

*Then*

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

**Example 2** (continued)
If $(A_n)$ is an increasing sequence, then the sequence of random variables $X_n = \mathbf{1}_{A_n}$ is also increasing. The Monotone convergence theorem then gives

$$\lim_{n\to+\infty} \mathbb{P}(A_n) = \lim_{n\to+\infty} \mathbb{E}[\mathbf{1}_{A_n}] = \mathbb{E}[\lim_{n\to+\infty} \mathbf{1}_{A_n}] = \mathbb{E}[\mathbf{1}_{\liminf A_n}] = \mathbb{E}[\mathbf{1}_{\cup_{n=1}^{\infty} A_n}] = \mathbb{P}(\cup_{n=1}^{\infty} A_n).$$

**Theorem 12.6** (Dominated Convergence Theorem)**.** *Let $(X_n)_{n=1}^{\infty}$ be a sequence of random variables such that*

$$X_n \to X \quad \textit{almost surely as } n \to \infty,$$

*and suppose there exists an integrable random variable $Y$ (i.e., $\mathbb{E}[|Y|] < \infty$) such that*

$$|X_n| \le Y \quad \textit{almost surely for all } n.$$

*Then*

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

*Proof.* Let $(X_n)_{n=1}^{\infty}$ be a sequence of random variables such that $X_n \to X$ almost surely, and suppose there exists an integrable function $Y$ with $|X_n| \le Y$ almost surely for all $n$. Since $0 \le Y + X_n \le 2Y$, we can apply Fatou's lemma to the nonnegative sequence $Y + X_n$:

$$\mathbb{E}[Y + X] = \mathbb{E}\Big[\liminf_{n\to\infty}(Y + X_n)\Big] \le \liminf_{n\to\infty} \mathbb{E}[Y + X_n] = \liminf_{n\to\infty} \left(\mathbb{E}[Y] + \mathbb{E}[X_n]\right),$$

which gives

$$\liminf_{n\to\infty} \mathbb{E}[X_n] \ge \mathbb{E}[X] - \mathbb{E}[Y] + \mathbb{E}[Y] = \mathbb{E}[X].$$

Similarly, apply Fatou's lemma to $Y - X_n \ge 0$:

$$\mathbb{E}[Y - X] \le \liminf_{n\to\infty} \mathbb{E}[Y - X_n] = \mathbb{E}[Y] - \limsup_{n\to\infty} \mathbb{E}[X_n],$$

which gives

$$\limsup_{n\to\infty} \mathbb{E}[X_n] \le \mathbb{E}[X].$$

Since

$$\liminf_{n\to\infty} \mathbb{E}[X_n] \le \limsup_{n\to\infty} \mathbb{E}[X_n],$$

we conclude

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

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**Theorem 12.7.** *Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables.*

1. *(Tonneli's theorem, discrete form) If $X_n \geq 0$ for any $n \in \mathbb{N}$, then*

$$\mathbb{E}\left[\sum_{n=1}^{+\infty} X_n\right] = \sum_{n=1}^{+\infty} \mathbb{E}[X_n]. \tag{12.2}$$

2. *(Fubini's theorem, discrete form) If $\sum_{n=1}^{+\infty} \mathbb{E}[|X_n|] < \infty$, then the series $\sum_{n=1}^{+\infty} X_n$ converges almost surely,* (12.2) *holds and both sides are finite.*

*Proof.* 1. Let $N \in \mathbb{N}$. By the linearity of expectation we have

$$\sum_{n=1}^{N} \mathbb{E}[X_n] = \mathbb{E}\left[\sum_{n=1}^{N} X_n\right].$$

We note that the sequence $(S_N = \sum_{n=1}^{N} X_n)_{N\in\mathbb{N}}$ is nonnegative and increasing, therefore, the Monotone Convergence Theorem as $N \to +\infty$ in the above equality gives

$$\sum_{n=1}^{+\infty} \mathbb{E}[X_n] = \lim_{N\to+\infty} \mathbb{E}\left[\sum_{n=1}^{N} X_n\right] = \mathbb{E}\left[\lim_{N\to+\infty} \sum_{n=1}^{N} X_n\right] = \mathbb{E}\left[\sum_{n=1}^{+\infty} X_n\right].$$

2. Note that by Tonelli's theorem, the condition $\sum_{n=1}^{+\infty} \mathbb{E}[|X_n|] < \infty$ yields

$$\mathbb{E}\left[\sum_{n=1}^{+\infty} |X_n|\right] < \infty.$$

Lemma 12.2 then implies $\sum_{n=1}^{+\infty} |X_n| < \infty$ almost surely, which gives (fact from real analysis) $\sum_{n=1}^{+\infty} X_n < \infty$ almost surely.
To prove (12.2) in this case, instead of using the MCT, we use the DCT with dominating random variable $Y = \sum_{n=1}^{+\infty} |X_n|$. We skip the details.

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